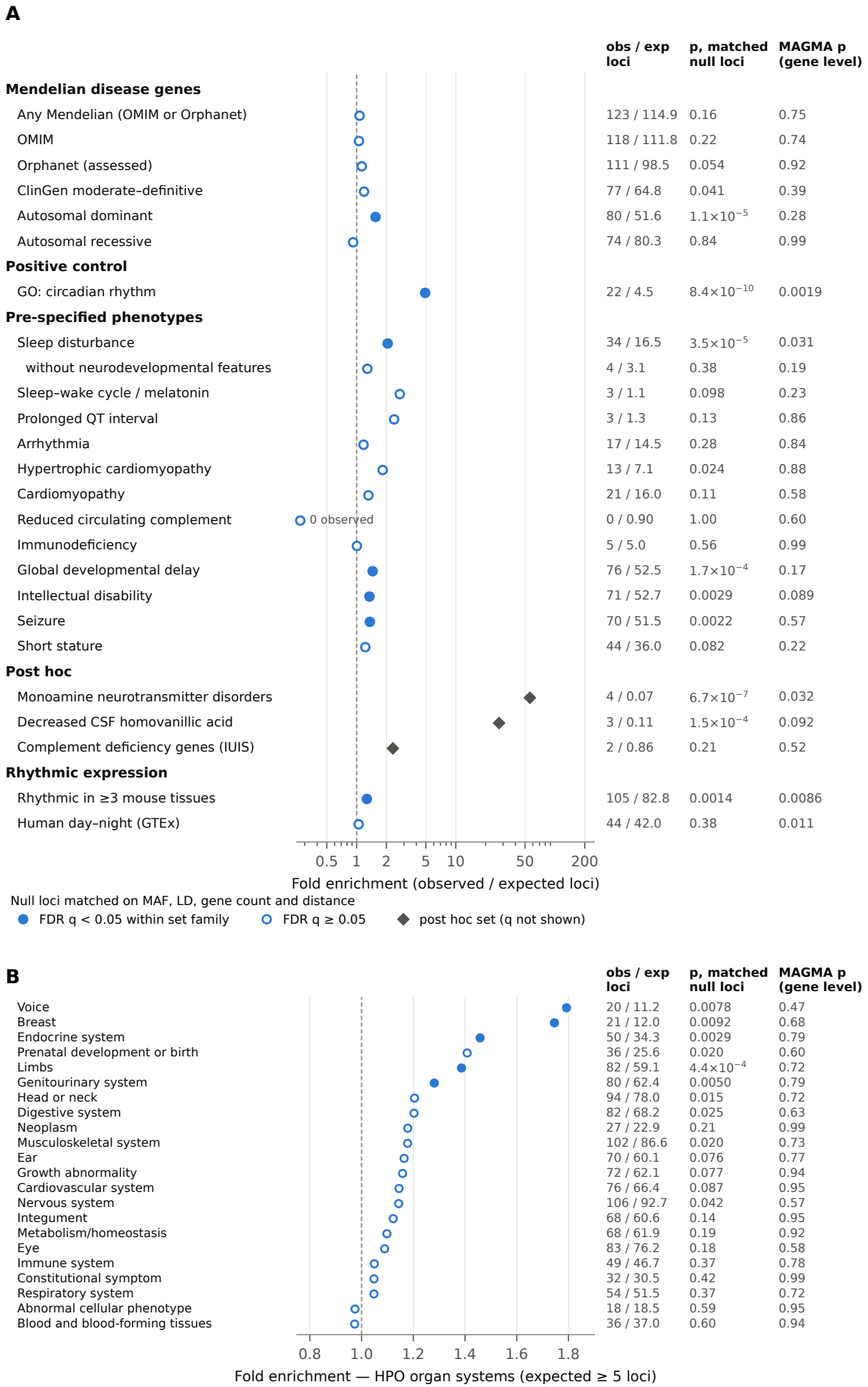
A
obs / exp loci
p, matched null loci
MAGMA p (gene level)
Mendelian disease genes
Any Mendelian (OMIM or Orphanet) 123 / 114.9 0.16 0.75
OMIM 118 / 111.8 0.22 0.74
Orphanet (assessed) 111 / 98.5 0.054 0.92
ClinGen moderate–definitive 77 / 64.8 0.041 0.39
Autosomal dominant 80 / 51.6 1.1×10−5 0.28
Autosomal recessive 74 / 80.3 0.84 0.99
Positive control
GO: circadian rhythm 22 / 4.5 8.4×10−10 0.0019
Pre-specified phenotypes
Sleep disturbance 34 / 16.5 3.5×10−5 0.031
without neurodevelopmental features 4 / 3.1 0.38 0.19
Sleep–wake cycle / melatonin 3 / 1.1 0.098 0.23
Prolonged QT interval 3 / 1.3 0.13 0.86
Arrhythmia 17 / 14.5 0.28 0.84
Hypertrophic cardiomyopathy 13 / 7.1 0.024 0.88
Cardiomyopathy 21 / 16.0 0.11 0.58
Reduced circulating complement 0 observed 0 / 0.90 1.00 0.60
Immunodeficiency 5 / 5.0 0.56 0.99
Global developmental delay 76 / 52.5 1.7×10−4 0.17
Intellectual disability 71 / 52.7 0.0029 0.089
Seizure 70 / 51.5 0.0022 0.57
Short stature 44 / 36.0 0.082 0.22
Post hoc
Monoamine neurotransmitter disorders 4 / 0.07 6.7×10−7 0.032
Decreased CSF homovanillic acid 3 / 0.11 1.5×10−4 0.092
Complement deficiency genes (IUIS) 2 / 0.86 0.21 0.52
Rhythmic expression
Rhythmic in ≥3 mouse tissues 105 / 82.8 0.0014 0.0086
Human day–night (GTEx) 44 / 42.0 0.38 0.011
0.5 1 2 5 10 50 200
Fold enrichment (observed / expected loci)
Null loci matched on MAF, LD, gene count and distance
FDR q < 0.05 within set family
FDR q ≥ 0.05
post hoc set (q not shown)
B
obs / exp loci
p, matched null loci
MAGMA p (gene level)
Voice 20 / 11.2 0.0078 0.47
Breast 21 / 12.0 0.0092 0.68
Endocrine system 50 / 34.3 0.0029 0.79
Prenatal development or birth 36 / 25.6 0.020 0.60
Limbs 82 / 59.1 4.4×10−4 0.72
Genitourinary system 80 / 62.4 0.0050 0.79
Head or neck 94 / 78.0 0.015 0.72
Digestive system 82 / 68.2 0.025 0.63
Neoplasm 27 / 22.9 0.21 0.99
Musculoskeletal system 102 / 86.6 0.020 0.73
Ear 70 / 60.1 0.076 0.77
Growth abnormality 72 / 62.1 0.077 0.94
Cardiovascular system 76 / 66.4 0.087 0.95
Nervous system 106 / 92.7 0.042 0.57
Integument 68 / 60.6 0.14 0.95
Metabolism/homeostasis 68 / 61.9 0.19 0.92
Eye 83 / 76.2 0.18 0.58
Immune system 49 / 46.7 0.37 0.78
Constitutional symptom 32 / 30.5 0.42 0.99
Respiratory system 54 / 51.5 0.37 0.72
Abnormal cellular phenotype 18 / 18.5 0.59 0.95
Blood and blood-forming tissues 36 / 37.0 0.60 0.94
0.8 1.0 1.2 1.4 1.6 1.8
Fold enrichment — HPO organ systems (expected ≥ 5 loci)